Passive Behavioral Tracing: A Non-Interventional Fairness Protocol for Cooperative AI

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Abstract

Emerging AI governance regimes, including the EU AI Act's post-market monitoring obligation and the NIST AI RMF's measure function, require ongoing fairness monitoring but leave open which statistics should satisfy it. Interventional audits (A/B tests, counterfactual procedures, adversarial probing) manipulate the system, and often the people, under audit, and are episodic where monitoring is continuous. This article argues that a useful part of that monitoring can be done passively, and proposes Passive Behavioral Tracing (PBT), a protocol that audits fairness in AI-mediated cooperative systems using only behavioral traces the system already records. PBT formalizes four constructs computable from ordinary logs (consistency of treatment, parity across naturally occurring groups, proportionality to contribution, and drift over time) and combines them into two indicators, PBT-Ethics and PBT-Coop. A Monte Carlo study with released code shows that the constructs form two families: reciprocity and parity register unfair treatment whether it has always existed or began recently, while consistency and drift respond to change, and only drift locates when the change happened. This distinction carries regulatory weight: unfairness present since deployment implicates a system's design, while unfairness with an onset points to a dateable change, and the two call for remedies that a single score cannot separate. The study also shows the protocol's limits: mild, sustained unfairness against a small group is often missed. All results come from a stylized synthetic model; PBT is a specification validated in principle and complementary to interventional audits, not a finding about any operating system.

Keywords: Algorithmic auditing; AI governance; Fairness monitoring; Post-market monitoring; Non-interventional methods; Cooperative AI

1 Introduction

Emerging AI governance regimes increasingly require not a one-time certification but ongoing fairness monitoring of deployed systems. The EU AI Act obliges providers of high-risk systems to operate a post-market monitoring system across the system's lifetime (European Union, 2024); the NIST AI Risk Management Framework's "measure" function calls for AI risks to be tracked over time rather than assessed once at deployment (National Institute of Standards and Technology, 2023); and the Digital Services Act requires very large platforms to conduct recurring systemic-risk assessments covering, among other things, discriminatory effects on users (European Union, 2022). Surveys of the AI ethics

guideline landscape find broad convergence on fairness and accountability as principles, alongside persistent silence on the procedural question of how compliance with them is to be established in practice (Jobin et al., 2019).

That silence is the problem this paper addresses. A regulator can require continuous fairness monitoring; what it cannot presently specify is which statistics an operator should compute, on what data, to satisfy the requirement, and this is not a gap that the existing fairness-auditing toolkit closes cleanly. That toolkit has largely inherited the experimental methods of the sciences that produced it: to know whether a system treats otherwise-similar people differently, audit methods typically manipulate something—the input a user is shown, the value of a protected attribute in a causal model, or the adversarial probes fed to the system—and observe the resulting difference in output. A/B testing exposes different users, often without their specific knowledge of the variant they received, to different system behavior in order to estimate a causal effect. Counterfactual-fairness procedures intervene on a causal graph, asking what a decision would have been had a protected attribute been different, which requires either a manipulated dataset or a simulated intervention on the model itself. Red-teaming and adversarial probing inject inputs designed to elicit a failure the system would not otherwise have produced in ordinary use.

There is a structural mismatch here that has received less attention than it deserves. A monitoring obligation is, by its nature, continuous and open-ended: it asks an operator to keep watching. An interventional audit is, by its nature, episodic and costly: it asks an operator to run an experiment. An operator told to monitor its high-risk system continuously for fairness, using methods that require manipulating live users to produce a finding, faces a choice between monitoring rarely enough to keep the manipulation tolerable and monitoring often enough to satisfy the obligation. Neither horn of that dilemma serves the people the obligation exists to protect.

These methods are valuable and, in the specific sense of supporting causal claims, often necessary. But the manipulation they require is not ethically free. Exposing live users to a deliberately varied treatment without full disclosure raises consent questions of its own; repeated experimental manipulation of a deployed system imposes a burden on the people subjected to it; and the very act of auditing can contaminate the system under test, since a system audited by injected probes is, during the audit, no longer behaving exactly as it would in ordinary operation. A fairness-auditing regime that itself depends on manipulating people or systems sits uneasily alongside the fairness commitments it is meant to verify. For a governance audience in particular, this creates a recurring dilemma: the audit method most capable of supporting a strong causal claim of unfairness is also the audit method least compatible with the consent and non-manipulation norms that motivated the fairness inquiry in the first place.

This paper asks whether fairness and cooperative behavior in an AI-mediated system can be audited using only the behavior the system and its users already produce, without any researcher-injected variation. We call the resulting protocol *Passive Behavioral Tracing* (PBT): a method for tracing, aggregating, and statistically evaluating naturally occurring behavioral traces—sequences of decisions, responses, and interactions that would have occurred whether or not an audit was taking place—in order to surface fairness and cooperation properties without intervention. Stated in plain terms, PBT asks four ordinary-language questions of a system's ordinary logs: does it treat the same person the same way twice in similar circumstances; does it treat two naturally occurring groups of people alike when their circumstances match; does its response track what a person actually put into the system; and does any of this quietly change over time without anyone noticing. Each question is given a precise statistical form

below, but none of them requires an auditor to introduce a single artificial condition that the system, or the people using it, would not otherwise have encountered.

PBT is deliberately positioned as the behavioral counterpart to a structural framework developed in a companion manuscript currently under review, in which two structural indicators (referred to here as Ripple-Flow and Ripple-Impact) evaluate the fairness of a cooperative-propagation structure's topology—who reciprocates with whom, and how benefit is distributed across a network. That structural framework is itself built on the structural-virality tradition in cascade research, which established that the shape of a propagation network, not merely its reach, carries information relevant to how a cascade should be evaluated. If Ripple asks whether the *structure* of cooperation is fair, PBT asks whether the *behavior* of the system toward individual participants, traced over time, is fair. As we show in Section 5, these two questions can have opposite answers for the same system, which is the central reason both are needed: a governance process that certifies a platform using only one of the two audits can certify a system that is failing badly along the dimension it never checked.

The systems this paper is concerned with are not a narrow technical niche. Mutual-aid platforms, time-banking systems, algorithmically mediated referral networks, and resource-allocation tools used by community organizations and public agencies share a basic shape: an AI-mediated layer decides, repeatedly and over time, how one participant's action is met with a response and how benefit moves between participants. These are precisely the systems for which the interventional toolkit is least comfortable to apply at scale. A public housing waitlist algorithm or a disaster-relief matcher cannot easily randomize who receives help without raising the consent and burden objections described above, yet these are also the systems whose fairness properties matter most to the people who depend on them.

The argument of this article can be stated in one sentence: for systems that mediate cooperation among many people over time, a meaningful part of the continuous fairness monitoring that regulation now demands can be carried out without manipulating anyone, provided the monitoring asks not only *whether* treatment is unfair but also *when* it became so. The contribution is a protocol rather than a new fairness definition: four constructs that can be computed from records an operator already keeps, an account of what each can and cannot detect, and released simulation code with which any reader can check those claims.

The remainder of the paper proceeds as follows. Section 2 situates PBT against interventional audit methods, existing fairness metrics, and the monitoring obligations of current AI governance frameworks. Section 3 gives formal definitions, with plain-language readings, of the four constructs PBT is built on. Section 4 develops these into ethical indicators, PBT-Ethics and PBT-Coop, discusses how a governance body might set thresholds for them in practice, and states the framework's causal and epistemic limitations. Section 5 presents a four-failure-mode taxonomy combining structural (Ripple) and behavioral (PBT) fairness, with a worked illustration of the failure mode a structural-only audit is built not to see. Section 6 develops the implications for AI governance, including why the distinction between unfairness that has an onset and unfairness present since inception carries regulatory weight, and states plainly what the evidence assembled here does and does not establish. Section 7 concludes.

2 Background: PBT and Interventional Auditing

2.1 Interventional fairness audits

Three interventional traditions dominate current fairness-auditing practice. A/B and multivariate testing randomizes users or requests across system variants and compares outcomes, providing strong causal identification at the cost of deliberately treating otherwise-similar people differently for the purpose of the test itself (Kohavi et al., 2020). Counterfactual-fairness approaches formalize fairness as invariance of a decision to a hypothetical change in a protected attribute within a specified causal model (Kusner et al., 2017), which requires either constructing counterfactual instances that manipulate that attribute or intervening directly on the model's causal structure (Pearl, 2009); the causal clarity these approaches offer depends on the correctness of the assumed causal model and on the manipulation being well-defined and consented to. Red-teaming and adversarial probing intentionally construct inputs designed to surface failure modes that would not arise under ordinary use (Raji and Buolamwini, 2019), which is valuable for discovering worst-case behavior but again departs from the system's naturally occurring behavior by construction.

Each of these traditions buys causal strength by spending the very thing a fairness audit is meant to protect: the ordinary, unmanipulated experience of the people the system serves. That trade-off is a structural feature of interventional auditing as such, and it is the one PBT is designed to avoid, at a cost this paper states explicitly.

2.2 Existing fairness metrics and their scope

Widely used fairness metrics—parity of outcomes across groups, equality of error rates, and related criteria—were developed primarily for classification and recommendation settings with a fixed decision at a fixed time (Hardt et al., 2016; Dwork et al., 2012; Chouldechova, 2017; Corbett-Davies and Goel, 2018; Zafar et al., 2017). They are not designed to characterize *propagation*: how a cooperative act moves through a network of interactions over time, or how a system's treatment of a participant evolves across repeated encounters. Auditing cooperative, propagation-based systems requires constructs computable over a behavioral trace rather than a single decision, extending the structural-virality tradition in cascade research, which established that the shape of a propagation network, not merely its reach, bears on how a cascade should be evaluated (Goel et al., 2016).

2.3 Monitoring obligations in AI governance frameworks

The regulatory instruments cited above converge on a requirement for ongoing observation but diverge, or remain silent, on its operationalization. The EU AI Act's post-market monitoring provision obliges high-risk providers to collect and analyze data on system performance throughout the deployment lifetime, without prescribing which fairness statistics that analysis should compute (European Union, 2024). The NIST AI RMF's "measure" function similarly directs organizations to track identified risks using quantitative and qualitative methods, leaving the choice of method to the organization (National Institute of Standards and Technology, 2023). The DSA's systemic-risk provisions require very large platforms to assess discriminatory effects periodically, again without specifying the statistical form such an assessment should take (European Union, 2022). This under-specification is defensible as regulatory design:

the appropriate statistic depends on the system, and premature standardization on a metric invites gaming. But it leaves operators and oversight bodies negotiating, case by case, over what evidence counts, and it leaves a specific gap for systems whose function is to mediate cooperation among many participants rather than to classify one person at a time.

Empirical work on how algorithmic accountability requirements are actually met in practice has documented a recurring pattern in which auditing obligations are satisfied through documentation and process rather than through measurement of a system's behavior toward its users (Ananny and Crawford, 2018; Kaminski, 2019). Our aim is not to displace procedural accountability, which we return to in Section 6, but to enlarge the set of behavioral measurements an oversight body can request without asking an operator to experiment on the people it serves.

2.4 Where PBT sits

PBT fills the resulting gap: constructs computable purely from the trace a system already produces in ordinary operation, requiring no injected variation, no causal intervention, and no adversarial probing. This buys ethical advantages—no additional manipulation of users, no contamination of the system under test, continuous rather than episodic applicability—at a stated cost in causal identification strength, which Section 4.4 makes explicit. Non-interventional auditing of algorithmic systems is not without precedent: black-box auditing methods have been developed that probe a deployed model's behavior using only its existing input-output surface rather than an experimentally manipulated one (Adler et al., 2018; Sandvig et al., 2014; Angwin et al., 2016; Bird et al., 2020), and quasi-experimental methods in econometrics have long extracted causal-adjacent estimates from naturally occurring variation rather than randomized assignment (Angrist and Pischke, 2009). PBT extends this non-interventional lineage from single-decision auditing and cross-sectional natural experiments to the specific setting of a behavioral trace observed over time in a cooperative-propagation system, which is where Trace Consistency and Drift Detection add constructs that the existing black-box and quasi-experimental literatures do not provide (Page, 1954; Basseville and Nikiforov, 1993). The motivation for seeking a non-interventional alternative in the first place is reinforced by two further adjacent literatures: sociotechnical accounts of fairness auditing that caution against treating any single formal test, interventional or not, as a self-sufficient guarantee of fair treatment (Selbst et al., 2019; Green and Viljoen, 2020), and scholarship on manipulation and surveillance that treats the unconsented manipulation of user behavior as a harm in its own right, independent of whether that manipulation is undertaken for auditing purposes or for engagement (Susser et al., 2019; Calo, 2014; Nissenbaum, 2004; Zuboff, 2019). This last literature is worth underscoring for a policy audience: an audit regime that manipulates users to test for fairness inherits the same consent and autonomy concerns that motivate scrutiny of manipulative design, even when well-intentioned.

3 Formal Definitions

We use *passive* in a narrower sense than "observational." An observational study can still involve the auditor choosing what to sample, when to look, or which subpopulation to compare, and those choices are a form of intervention on the audit process even if the system is not manipulated. Passive tracing additionally requires that the variation used for comparison—across contexts, groups, and time—already

existed in the system's ordinary operation before the audit began, so that the auditor's only role is to record and aggregate a trace that would have been produced anyway. This is what licenses the term *non-interventional*: passivity is a property of the relationship between the audit and the system's behavior, not merely a description of the auditor's data-collection style.

Let a system produce, over an observation window, a stream of interaction records $\{(u_t, x_t, y_t)\}_{t=1}^{T}$, where u_t identifies the participant involved in interaction t, x_t is the naturally occurring context of that interaction (unmanipulated by the auditor), and y_t is the system's resulting behavior toward u_t (a decision, an allocation, a response). Crucially, x_t is drawn from whatever variation already exists in ordinary use—different participants, different times, different naturally occurring contexts—rather than variation assigned by an auditor. The four constructs below are all computed from this stream alone.

Stated as a data requirement, PBT needs four fields: a participant identifier, the context of each interaction, the system's response, and a timestamp or interval index (required only by Drift Detection, which is the one construct that reads the trace's temporal order). These are fields that cooperative and allocation systems record as a matter of ordinary operation, because each is needed to run the system at all rather than to audit it. A housing-allocation engine, a disaster-relief matcher, or a referral platform already logs who applied, under what circumstances, what the system returned, and when. PBT therefore imposes no additional data-collection burden and requires no new instrumentation, which is the practical basis for the governance claims developed in Section 6.

3.1 Trace Consistency

Trace Consistency, $TC(u)$, measures how stably a system treats a given participant u across the naturally occurring repeated instances of similar context in u's own trace, without any injected repetition:

$$TC(u) = 1 - \frac{1}{|P_u|} \sum_{(t,t') \in P_u} \frac{|y_t - y_{t'}|}{\max(y_t, y_{t'}) + \epsilon}, \tag{1}$$

where P_u is the set of naturally occurring pairs of interactions involving u under sufficiently similar context $x_t \approx x_{t'}$ (determined by a similarity threshold on pre-existing context variation, not by researcher-assigned repetition), and $\epsilon > 0$ avoids division by zero. High $TC(u)$ indicates the system behaves consistently toward u across the naturally occurring variation in u's own history.

In plain terms. $TC(u)$ asks: the next time person u is in roughly the same situation as before, does the system respond the same way? The auditor never places u in that situation twice on purpose; the situation recurs naturally over the life of an ongoing platform, and the two responses are compared.

3.2 Non-Interventional Fairness

Non-Interventional Fairness, NIF, adapts a conditional-independence fairness criterion to naturally occurring covariate variation rather than researcher-assigned variation. Let $g(u)$ denote a naturally occurring grouping of participants (e.g., by a pre-existing, non-manipulated attribute) and let $Y \mid X, g$ denote the distribution of system behavior conditional on context and group, estimated only from the

variation already present in the trace:

$$NIF = 1 - \sup_{x} \; d\big(Y \mid X{=}x, g{=}g_1, \;\; Y \mid X{=}x, g{=}g_2\big), \tag{2}$$

where $d(\cdot, \cdot)$ is a distributional distance (e.g., total variation) between the behavior distributions of two naturally occurring groups matched on context x, and the supremum is taken over contexts with sufficient naturally occurring support in both groups. NIF is high when, across the contexts naturally shared by both groups, the system's behavior does not differ.

In plain terms. NIF takes two groups that already exist in the population, for instance, people who joined a platform early versus late, and looks only at the situations both groups have naturally already encountered. Within those shared, unmanipulated situations, does the system behave the same way toward both groups? No group is assigned to a condition by the auditor; the comparison is confined to overlap that already exists in ordinary use.

3.3 Behavioral Reciprocity

Behavioral Reciprocity extends the reciprocity concept from structural analysis to AI-mediated behavior: it asks whether the system's treatment of a participant responds proportionately to that participant's own prior contribution, as observed passively rather than manipulated. Writing c_t for participant u_t's naturally occurring contribution level at interaction t (e.g., prior cooperative input to the system) and y_t for the system's resulting behavior toward them,

$$BR(u) = \mathrm{corr}\big(\{c_t\}_{t \in u}, \{y_t\}_{t \in u}\big), \tag{3}$$

the correlation, over u's naturally occurring trace, between contribution and resulting treatment. High $BR(u)$ indicates the system's treatment of u tracks u's own contribution rather than being unresponsive or inversely related to it.

In plain terms. $BR(u)$ is a proportionality check: when a person puts more into a cooperative system, does what they get back scale up accordingly? A system that ignores contribution, or rewards it less the more a person gives, scores low here, using nothing but the ordinary record of what people contributed and received.

3.4 Drift Detection

Drift Detection identifies gradual change in any of the above quantities over time using only the sequential structure already present in the trace, without intervention. For a quantity Q_t (e.g., a windowed estimate of NIF or BR) computed over successive naturally occurring intervals, we apply a sequential change-point procedure to $\{Q_t\}$ to flag intervals in which the quantity shifts beyond what naturally occurring noise would predict, using only the passively observed sequence rather than a designed before/after comparison. The procedure's reference level is the system's own behavior at the beginning of the observation window, which is what makes Drift Detection a detector of *change* rather than of a level

of unfairness; the instantiation used in this article is a two-sided cumulative-sum (CUSUM) procedure (Page, 1954; Basseville and Nikiforov, 1993), specified in Section 4.7.

In plain terms. Behavioral Reciprocity and Non-Interventional Fairness summarize a trace as a whole, so they register that a group is treated worse but cannot say since when. Trace Consistency notices that a person's treatment has changed, because the same situation is met with different responses at different times, but it does not say when the change occurred. Drift Detection watches the sequence over time and flags the interval, if there is one, from which behavior toward a group differs from how it was treated at the start of the observation window.

4 Ethical Indicators: PBT-Ethics and PBT-Coop, and Their Limits

4.1 PBT-Ethics

We define $PBT\text{-}Ethics$ as a combination of NIF and drift-adjusted consistency, aggregated across the participant population:

$$PBT\text{-}Ethics = NIF \cdot \big(1 - \delta_{\text{drift}}\big), \tag{4}$$

where δ_{drift} is the fraction of observation intervals flagged by Drift Detection. This indicator is high when the system's behavior is both non-interventionally fair across naturally occurring groups and stable over time, without requiring any manipulation to establish either property.

4.2 PBT-Coop

We define $PBT\text{-}Coop$ as the population-average Behavioral Reciprocity, weighted by Trace Consistency:

$$PBT\text{-}Coop = \frac{1}{|U|} \sum_{u \in U} TC(u) \cdot BR(u). \tag{5}$$

This indicator is high when the system's treatment of participants tracks their own cooperative contribution consistently across their naturally occurring history.

In plain terms. $PBT\text{-}Ethics$ answers "is this system fair across groups, and has that fairness held steady over time?" $PBT\text{-}Coop$ answers a related but distinct question: "does this system reward cooperation proportionately, and consistently for the same person?" A platform can score well on one and poorly on the other, which is itself diagnostic information a single combined score would hide.

4.3 Theoretical connection

PBT-Ethics targets a non-manipulation-respecting form of equal treatment: fairness established without the auditor exposing participants to conditions they would not otherwise have encountered, which speaks directly to concerns about participant autonomy and burden that interventional audits raise. PBT-Coop targets whether cooperative behavior is met with proportionate treatment over time, a behavioral analogue of the reciprocity concern raised structurally by Ripple-Flow in the companion manuscript, but assessed at the level of individual behavioral history rather than network topology.

4.4 Limitations

Three limitations must be stated without qualification. First, non-intervention costs causal identification: because x_t and $g(u)$ are not randomized, NIF and BR remain vulnerable to unmeasured confounding in a way a randomized intervention would not be; a high NIF establishes conditional behavioral parity given the observed context, not a causal fairness guarantee in the counterfactual sense interventional methods target. Second, behavioral logs reflect what the system did and what was recorded, not participants' experience of the interaction; a system can score well on every construct here while participants feel poorly treated in ways the trace does not capture. Third, as with the Ripple indicators in the companion manuscript, the functional forms chosen are one reasonable operationalization among several, and any deployment should treat the form as a design decision requiring independent justification. A governance body adopting PBT should read a high score as evidence of the absence of the specific failures these constructs detect, not as a general certificate of fair treatment.

4.5 Setting Thresholds in Practice

None of the constructs above is useful to a governance process without a way to decide how high NIF must be, how large a distributional distance counts as a violation, or how sensitive Drift Detection should be. We do not propose a universal threshold, for the reason Section 4.4 gives for not fixing the functional forms: the right threshold is a context-dependent governance decision, not a technical constant. We can, however, specify how that decision should be made. Three approaches are available and are not mutually exclusive. First, a threshold can be calibrated against a system's own historical baseline: rather than fixing $NIF \geq 0.9$ as a universal bar, an operator can require that NIF not fall below the level observed over a prior reference period. Instantiated concretely, an operator might take the mean NIF over the preceding twelve months as its reference and treat a fall of more than some stated proportion below that mean as a fairness degradation requiring investigation. We give the shape of such a rule rather than a number, because the tolerable proportion depends on the system's own variability and on what is at stake for its participants; the point of the baseline approach is that the bar is derived from the system's own record rather than asserted. Second, thresholds can be set through participatory or regulatory processes in which the affected populations, or a body accountable to them, specify what distributional distance or drift magnitude they are willing to accept, analogous to tolerance bands in statistical quality control. Third, and most tractable absent either historical data or a participatory process, an operator can report NIF, BR, and drift statistics across a range of candidate thresholds—a sensitivity analysis—so that a governance body sees how the verdict would change under different choices rather than receiving a single pass/fail bit computed at an unexplained cutoff. We adopt the third approach below, to make the threshold dependency visible rather than hidden inside a reported number. For a regulator without access to a platform's historical data, this approach has a further advantage: it can be requested and reviewed without first agreeing on a universal numerical bar, often the point at which cross-jurisdictional standard-setting stalls.

4.6 A Worked Numeric Illustration

To make the constructs concrete, consider a synthetic trace for a single participant u observed over 12 naturally occurring intervals, with contribution levels ranging from 2 to 9 on an arbitrary scale and

corresponding system responses ranging from 3 to 11. Behavioral Reciprocity, the correlation between the two sequences, gives $BR(u) \approx 0.87$: the system's treatment of u tracks u's contribution closely. Trace Consistency, computed over the 4 pairs of intervals in which u's context was most similar, gives $TC(u) \approx 0.81$. For Non-Interventional Fairness, take two naturally occurring groups, participants who joined during the platform's first month versus later joiners, matched on context; if the maximum distributional distance between their treatment over contexts with adequate support in both is $d = 0.06$, then $NIF = 0.94$. If Drift Detection flags 1 of 12 intervals, $\delta_{\text{drift}} \approx 0.083$ and $PBT\text{-}Ethics \approx 0.86$. Reporting that figure against the three thresholding approaches above—a historical baseline of 0.80 from the prior quarter, a participatory tolerance band of $[0.75, 1.0]$, and a sensitivity sweep from 0.70 to 0.95—shows it passing under all three, which is the transparency about the threshold decision that a single unexplained pass/fail bit would obscure. As in the companion structural framework, these numbers are an illustrative construction, not a measurement of any deployed system.

4.7 Synthetic Validation

The illustration above uses a single hand-constructed instance to make the constructs concrete. We now report a Monte Carlo validation across many independently simulated populations, to check that the constructs behave as claimed rather than only in one hand-picked case. The simulation code is released with the article, and every number in this subsection is produced by it.

Design. We simulate a platform with 80 main-population participants and a 20-participant subgroup over $T = 12$ intervals, one interaction per participant per interval. Each interaction occurs in one of four naturally occurring contexts with a context effect $\kappa_x \in \{0, 0.3, 0.6, 0.9\}$, and each participant's contribution c_t is drawn from a Gamma distribution with shape 2 and scale 2. Under fair treatment the system's response is $y_t = 1 + c_t + \kappa_x + \varepsilon_t$ with $\varepsilon_t \sim \mathcal{N}(0, 0.5^2)$; under unfair treatment the subgroup's response is weakly coupled to contribution and suppressed in level, $y_t = 0.6\,(1 + 0.35\,c_t + \kappa_x) + \varepsilon_t$. Responses are floored at 0.05, so that every construct stays within its stated range. Three conditions are compared: *fair* (the subgroup is treated like everyone else), *sustained-unfair* (the unfair rule applies to the subgroup in all T intervals), and *onset-unfair* (the subgroup is treated fairly for the first six intervals and unfairly from the seventh). Each condition is repeated for $1{,}000$ independently sampled populations.

NIF uses total-variation distance on five response bins (pooled quantiles), with the supremum taken over contexts in which each group has at least 20 interactions. For Drift Detection, Q_t is the subgroup's mean response in interval t divided by the mean response that the main group's own rule (a least-squares fit of response on contribution and context within that interval) would predict for the subgroup's contributions and contexts; $Q_t = 1$ means the subgroup is treated like everyone else. A two-sided CUSUM with reference value 0.5 is run on Q_t standardized by the mean and standard deviation of the first four intervals, and its alarm threshold ($h = 11.5$) is calibrated on $2{,}000$ separate fair runs so that a fair series raises any alarm in 5% of runs. δ_{drift} is the fraction of the 12 intervals flagged.

Two families of constructs. Table 1 and Figure 1 show that the four constructs do not respond to unfairness in the same way. Behavioral Reciprocity and Non-Interventional Fairness fall under both unfair conditions: the subgroup's treatment no longer tracks its contribution, and its response distribution separates from the main group's within shared contexts. Trace Consistency behaves differently. Under

	Fair	Sustained-unfair	Onset-unfair
$BR(u)$ (subgroup mean)	0.971 ± 0.005	0.704 ± 0.041	0.536 ± 0.061
NIF	0.839 ± 0.036	0.212 ± 0.028	0.550 ± 0.040
$TC(u)$ (subgroup mean)	0.605 ± 0.016	0.617 ± 0.019	0.485 ± 0.020
δ_{drift}	0.019 ± 0.080	0.013 ± 0.066	$\mathbf{0.500 \pm 0.008}$
Runs with any drift alarm	7.2%	4.7%	100.0%
$PBT\text{-}Ethics$	0.823 ± 0.076	0.209 ± 0.031	0.275 ± 0.020
$PBT\text{-}Coop$ (whole population)	0.586 ± 0.007	0.555 ± 0.008	0.519 ± 0.008

Table 1: Monte Carlo validation, 1,000 simulated populations per condition (mean ± SD). BR and NIF fall under both forms of unfairness. TC and δ_{drift} stay at their fair-condition level under sustained unfairness, because the subgroup is treated badly but consistently, and move only when treatment changes partway through the trace. Of the four, only Drift Detection also locates the change in time.

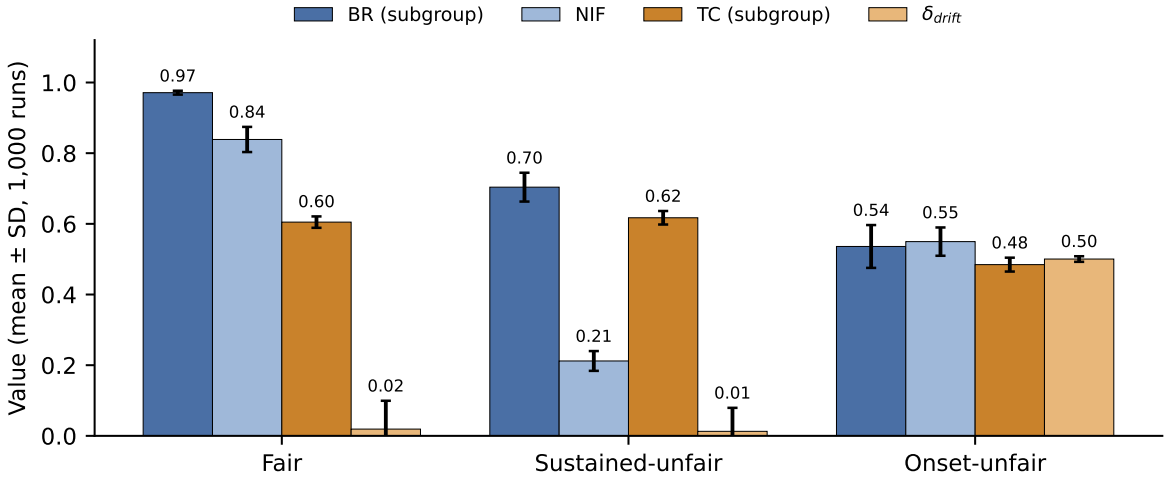


Figure 1: The four constructs under the three conditions (mean ± SD over 1,000 runs). Reciprocity and parity respond to the *level* of unfairness; consistency and drift respond to *change*.

sustained unfairness it is indistinguishable from the fair condition (0.62 against 0.60), because a subgroup that is treated badly but in the same way every time is, by this measure, treated consistently. It falls only under onset unfairness (0.48), where the same situation is met with different responses before and after the change. Drift Detection is the sharpest form of this second family: its alarm rate under sustained unfairness (4.7%) is no higher than under fair treatment (7.2%, against a nominal 5%), because there is no change for it to detect, while under onset unfairness it raises an alarm in 100% of runs, flags 0.50 of the intervals (the six post-onset intervals are half of twelve), and in only 0.5% of runs flags an interval before the true onset. The median delay between onset and first alarm is 0 intervals.

The practical reading is that the four constructs answer two different questions. BR and NIF answer "is this group treated worse?", and TC and Drift Detection answer "has treatment changed?". Only Drift Detection answers "since when?", which is the question whose governance consequences Section 6.1 develops. The two composite indicators inherit this split. $PBT\text{-}Ethics$, which multiplies NIF by the share of unflagged intervals, is low under both unfair conditions. $PBT\text{-}Coop$, averaged over the whole population, moves only slightly (0.586, 0.555, 0.519), because the 80 fairly treated participants dominate the average. A population-level reciprocity score can therefore hide a subgroup's

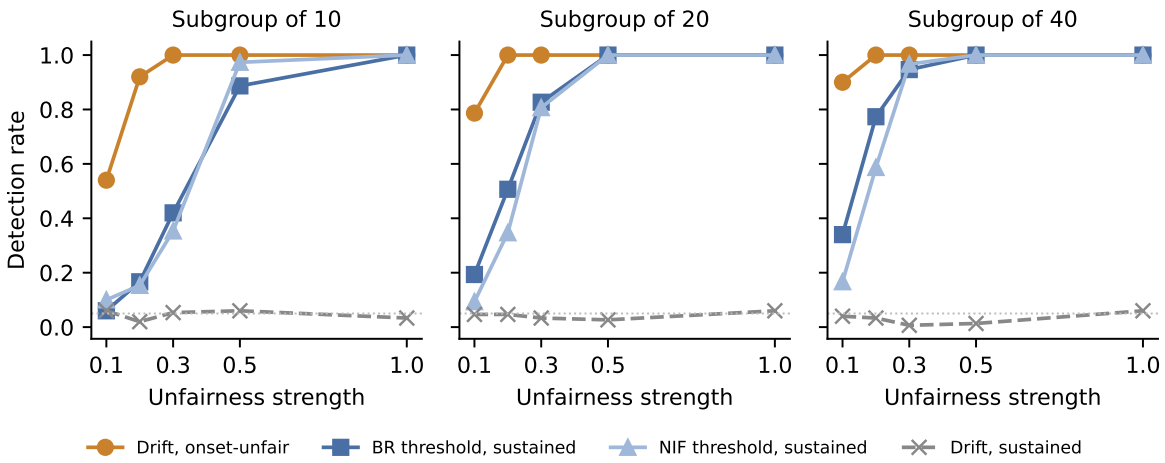


Figure 2: Detection rate against unfairness strength (0 = fair rule, 1 = the unfair rule of the main experiment) for subgroups of 10, 20 and 40 participants; 150 runs per point, thresholds calibrated on fair runs of the same subgroup size. The dotted line marks 5%. Drift Detection catches even mild onset unfairness, but threshold-based detection of mild *sustained* unfairness is unreliable for small groups.

treatment, and we recommend that $PBT\text{-}Coop$ always be reported per group as well as overall.

Detection with a calibrated threshold. Following the historical-baseline approach of Section 4.5, we set a threshold at the 5th percentile of the subgroup's BR in half of the fair runs and apply it to the other half and to the unfair runs. The threshold flags 100% of sustained-unfair and 100% of onset-unfair populations, and 4.4% of held-out fair populations.

Where the protocol fails. The unfair rule in the main experiment is a large effect, so the results above say little about weaker unfairness. We therefore interpolate between the fair and unfair rules with a strength parameter between 0.1 and 1 and vary the subgroup size (Figure 2). Drift Detection remains sensitive: at strength 0.1, the weakest tested, it detects onset unfairness in 54%, 79% and 90% of runs for subgroups of 10, 20 and 40, because each interval compares the subgroup with the main group's treatment of similar contributions at the same time. Detecting *sustained* unfairness through a calibrated BR or NIF threshold is much harder. At strength 0.2, the BR threshold catches 17% of affected populations when the subgroup has 10 members and 51% when it has 20; reliable detection (above 80%) requires strength 0.3 or more for a subgroup of 20. This is the protocol's most important weakness: mild unfairness that has been present from the start and affects a small group is exactly the case a passive audit is most likely to miss, because there is no change to detect and too few observations to separate a small level difference from noise. An oversight body relying on PBT should treat a clean result for a small group as weak evidence, and this is one of the places where an interventional audit retains a role PBT cannot fill.

5 Four Failure Modes: Structural and Behavioral Fairness Combined

Because Ripple-Flow/Ripple-Impact (structural fairness) and PBT-Ethics/PBT-Coop (behavioral fairness) are computed from different aspects of a system—topology of propagation versus individual be-

havioral trace over time—they can diverge. We describe four combinations.

High structural fairness, low behavioral fairness. A cooperative platform's aggregate propagation structure may be reciprocal and evenly distributed at the network level (high Ripple-Flow/Impact) while, at the level of individual behavioral treatment, the system's responses to specific participants drift over time or fail to track their actual contribution (low PBT-Coop, drift flagged). The structural snapshot looks healthy while individual participants experience inconsistent or unresponsive treatment that a topology-level metric cannot see.

Low structural fairness, high behavioral fairness. A structure organized around a small number of hubs (low Ripple-Flow, concentrated onward-giving) can nonetheless treat each individual participant it does interact with consistently and proportionately to their contribution (high PBT-Coop), and can maintain non-interventional parity across naturally occurring groups within the population it serves (high NIF). The structure is hierarchical, but behavioral treatment within it is not, for that reason, automatically unfair.

Both high. Structural reciprocity and even distribution are matched by consistent, proportionate, non-drifting individual treatment: the strongest case, and the one an operator should be aiming for.

Both low. Concentrated, extractive structure is compounded by inconsistent or unresponsive individual treatment and undetected drift: the most serious failure, in which neither the topology nor the individual behavioral record offers reassurance.

The upshot for a governance audience is that a process relying on structural audit alone, or behavioral audit alone, can certify a system failing badly on the dimension it did not check. A requirement written around only one audit type would, on its own terms, be satisfiable by a system sitting in either of the first two failure modes above.

A concrete illustration of the high-structure/low-behavior case. Take a referral platform whose aggregate propagation network scores $F(G) = 0.85$ and $I(G) = 0.79$ on the companion structural framework (matching the Case A/B constructions of that framework), so that a structural audit alone would certify the platform as fair on both relational and distributive grounds. Suppose, however, that a subset of long-tenured participants receives systematically slower or less proportionate responses as the platform scales—a pattern invisible in the aggregate topology because it involves only within-node behavior over time, not the shape of the propagation graph. Computed on this subgroup's individual traces, $BR(u)$ falls to around 0.31 for these participants against 0.87 for newer participants, and Drift Detection flags 5 of 12 observation intervals for exactly this subgroup, giving $\delta_{\text{drift}} \approx 0.42$ where the platform-wide figure (Section 4.6) was 0.083. A structural-only audit reports $F(G) = 0.85, I(G) = 0.79$ and stops; a combined audit additionally reports that a specific, identifiable subgroup is being treated with roughly a third of the reciprocal responsiveness afforded to everyone else, which is precisely the failure the structural numbers alone are built not to see.

6 Implications for AI Governance

PBT is best understood as a continuously deployable monitoring layer suited to governance contexts in which repeated experimental manipulation of live participants is itself the ethically costly step—high-stakes cooperative platforms, resource-allocation systems serving vulnerable populations, or any deployment where an operator is reluctant, for good reason, to run repeated A/B variants on real users.

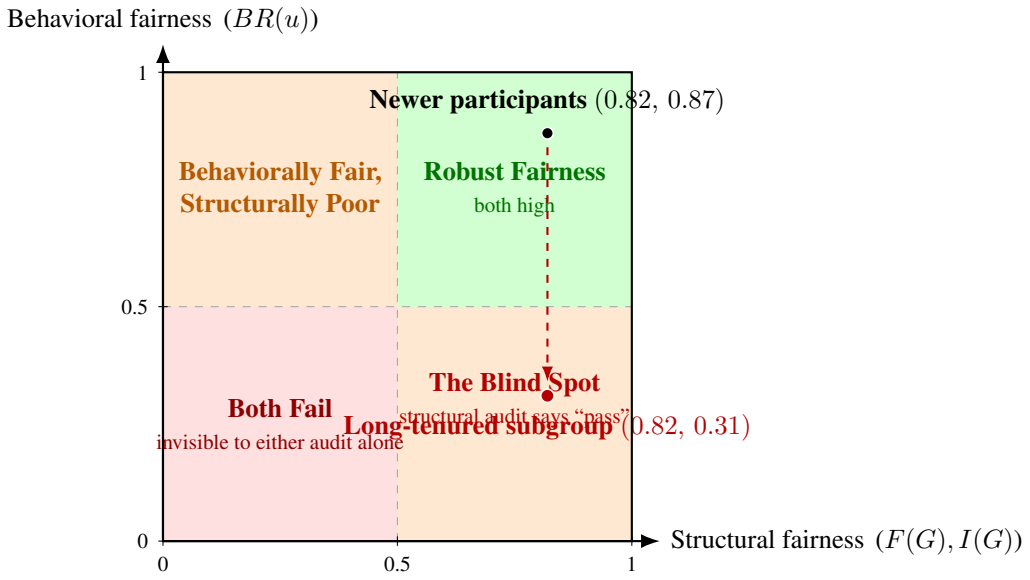


Figure 3: Same platform, same structural score ($F(G)$=0.85, $I(G)$=0.79), two very different behavioral realities. Newer participants sit in the Robust Fairness quadrant; a long-tenured subgroup, at identical structural coordinates, falls into the region a structural-only audit cannot see. The vertical drop between the two points *is* the failure Section 4 argues a structural metric alone cannot detect.

In such settings PBT allows an auditor to maintain an ongoing, passive read on fairness and reciprocity between the periodic interventional audits that remain necessary for stronger causal claims.

We do not propose PBT as a substitute for interventional causal audits where those are ethically and practically feasible; Section 4.4 is explicit that non-intervention weakens causal identification relative to randomization. Rather, PBT fills the gap between such audits, and is the appropriate tool precisely where an interventional audit's own ethical cost is what motivated seeking an alternative.

6.1 Why the onset/persistence distinction has regulatory weight

One result of the synthetic validation has a policy consequence that is easy to miss when it is stated statistically. A regulator or oversight body that receives a single passing or failing fairness score learns that a system is or is not treating a group unfairly. It does not learn *when* that treatment began, and the difference between the two cases is not a technical refinement but a difference in what the finding implies and what remedy it calls for.

A sustained failure, unfairness present since a system's inception, indicates a system that has never met the fairness commitment it was deployed under. The finding points at the system's original design and at whatever conformity assessment certified it, and the remedy is redesign. An onset failure, unfairness that began partway through the observation window, points instead at a specific, dateable change: a ranking-algorithm update, a policy revision, a moderation-rule change, a shift in the population being served. The finding localizes responsibility to an identifiable decision, and the remedy may be as narrow as reverting or revising that decision. These are different regulatory conclusions with different evidentiary implications, and a composite score reported alone cannot distinguish them.

Section 4.7 shows that the distinction is recoverable passively: BR and NIF register that a group is treated worse whether or not the treatment changed, TC registers that it changed but not when, and

Drift Detection separates the two cases and locates the onset, raising an alarm in every simulated onset case while staying at its false-alarm rate when unfairness was present from the start. A monitoring requirement written to ask for a reciprocity or parity statistic *alongside* a drift statistic would therefore yield strictly more actionable information than one asking for either alone, at no additional cost in data collection and no additional intervention on users. This is a concrete, implementable specification of the sort that Article 72 of the AI Act (European Union, 2024) and the NIST AI RMF's measure function (National Institute of Standards and Technology, 2023) currently leave open, and it is the most direct policy implication of the results reported here.

6.2 Evidence status of the results reported here

Because this paper's claims are addressed in part to a policy audience, the evidentiary basis for them should be stated plainly rather than left to be inferred. All quantitative results reported in Section 4 come from synthetic data generated by the stylized statistical model specified in Section 4.7; no deployed platform's data is used, and no claim is made about any deployed platform. What the simulation establishes is that the four constructs behave as designed under a known generative process, and specifically that Drift Detection separates onset from sustained unfairness and is the only construct that locates the onset in time. It also establishes a limit: mild sustained unfairness against a small group is often missed (Figure 2). What it does not establish is how the constructs behave on real behavioral logs, where the response function, the noise structure, and the mechanism generating unfairness may all differ from those assumed here in ways this paper cannot anticipate.

We therefore present PBT as a specification and a validated-in-principle proposal, not as an audited finding about any operating system. A policy recommendation that a monitoring regime request these statistics would, on the evidence assembled here, be a recommendation to try them, subject to the empirical validation described in Section 7 as future work. We regard stating this distinction explicitly as preferable to the alternative, common in framework proposals of this kind, of leaving readers to work out from the methods section how much of the claimed capability has actually been demonstrated.

6.3 A concrete monitoring workflow

The claims above remain abstract until it is clear what an operator would actually do. PBT can be run as a three-step cycle. First, *log collection*: the four fields identified in Section 3—participant identifier, context, system response, timestamp—are extracted from operational logs for a defined reporting period. Second, *indicator computation*: TC, NIF, BR, and Drift Detection are computed over that period and combined into $PBT\text{-}Ethics$ and $PBT\text{-}Coop$. Third, *reporting*: the operator submits the current values, their trajectory over the preceding reporting periods, any intervals flagged by Drift Detection, and the threshold rationale under which those flags were raised.

A regulator could request $PBT\text{-}Ethics$ and $PBT\text{-}Coop$ on a recurring basis as part of a post-market monitoring report, computed entirely from existing logs without any manipulation of users. We name no specific cadence, because none of the instruments discussed in Section 2.3 prescribes one and the appropriate interval depends on how quickly a system's behavior can change; a slow-moving allocation process and a continuously retrained ranking system do not need the same reporting frequency. What the results in Section 4.7 do imply is a constraint on how short the reporting window can be: Drift Detection

needs enough intervals within a window to locate a change point, so a window too short to contain the change is a window in which the onset/persistence distinction is not recoverable.

This suggests a two-layer arrangement when PBT is paired with the companion structural framework. Structural properties change slowly, as they reflect the topology a platform's design produces, and can reasonably be certified on a long cycle; behavioral treatment of individuals can shift between certifications, which is what Drift Detection is built to catch, and warrants the shorter cycle. A regime that certified structural fairness periodically using Ripple-Flow and Ripple-Impact while requiring more frequent behavioral reporting under PBT would close the gap Section 5 identifies, in which a system passing an audit on one axis fails undetectably on the other.

6.4 Relevance for governance and oversight practice

Two further features bear on how an oversight body might use PBT. Because the underlying logs are records an operator already retains, a body reviewing PBT-derived indicators is asking for a statistical summary of existing records rather than for a new experiment on users; this lowers the barrier to requesting an audit, though it does not remove the need to verify that the logs shown are complete and unaltered (Ananny and Crawford, 2018). And the threshold-setting approach of Section 4.5 supports external review: a sensitivity analysis across candidate thresholds lets a body evaluate an audit's conclusions without first agreeing with the operator on a universal numerical bar, frequently the point at which such negotiations stall.

None of this substitutes for the causal strength of a well-designed interventional audit where one is feasible and ethically warranted, and a governance regime that relied on PBT exclusively would inherit the confounding vulnerability described in Section 4.4. The two-layer picture argued for in Section 5, structural audit and behavioral audit run together, extends naturally to a three-layer picture for governance purposes: periodic interventional audits for causal strength, continuous structural monitoring for network-level fairness, and continuous behavioral monitoring via PBT for individual-level consistency and reciprocity, with each layer compensating for a specific limitation of the other two.

A caution belongs alongside this proposal rather than after it. Current governance frameworks lean toward documentation-based and procedural requirements, impact assessments, model cards, audit trails, partly because a single number is easy to game and easy to misinterpret without the context a procedural review supplies, and partly because transparency about a system's internals does not by itself yield accountability (Ananny and Crawford, 2018; Kaminski, 2019). PBT's outputs do not substitute for that context: a $PBT\text{-}Ethics$ score reported without the underlying trace, the population it covers, and the threshold rationale of Section 4.5 is exactly the unexplained pass/fail bit this paper argues against. The constructs are best read as inputs to a documentation-based review, evidence a reviewer can interrogate and recompute under alternative thresholds, not as an automated verdict replacing that review.

7 Conclusion

This paper has proposed Passive Behavioral Tracing as a protocol for auditing fairness and cooperative behavior in AI-mediated systems using only naturally occurring behavioral traces. We formalized four constructs—Trace Consistency, Non-Interventional Fairness, Behavioral Reciprocity, and Drift Detection—built entirely from logs platforms already keep, gave each a plain-language reading for read-

ers outside the technical literature, and combined them into two indicators, PBT-Ethics and PBT-Coop. A Monte Carlo study with released code showed that the constructs form two families: BR and NIF register that a group is treated worse regardless of when the treatment began, while TC and Drift Detection respond to change, and only Drift Detection locates the onset. That separation matters for governance because the two cases implicate different responsible decisions and call for different remedies. The same study showed where the protocol is weak: mild, sustained unfairness against a small group is often missed. We further argued, with a worked illustration, that behavioral and structural fairness are independent axes, producing four failure modes that a single-axis audit cannot surface, with direct consequences for how a monitoring requirement written around one audit type could be satisfied by a system that is nonetheless failing its users.

We have been explicit about what this evidence does not reach. Non-intervention weakens causal identification relative to randomized methods; behavioral logs remain a proxy for, not a measure of, participants' own experience of fairness; and every quantitative result here describes the constructs' behavior under a stylized generative model rather than on any deployed platform. Empirical instantiation on real logs and validation against participant-reported fairness are the necessary next steps, and until they are taken PBT should be read as a specification that a monitoring regime could adopt and test, not as a finding about any system now in operation.

Statements and Declarations

Funding. No external funding was received for conducting this study.

Competing interests. The author leads YOROSIX, a project developing a cooperative-behavior platform in the broad domain this paper's framework is intended to apply to. The synthetic validation (Section 4.7) uses simulated data from a stylized model specified in full in the paper, and makes no claim about data from any specific platform.

Data and code availability. No empirical data were used. The simulation code (`pbt_sim.py`) and its outputs, which reproduce every number in Section 4.7, are available at `https://github.com/kazutakahayashiyorosix/passive-behavioral-tracing`.

Author contributions. The sole author conceived the framework, developed the formal model and synthetic validation, and wrote the manuscript.

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